Structural basis of ZH853 recognition by the µ-opioid receptor: a distinctive extracellular-loop contact defines a selectivity handle and guides analog design

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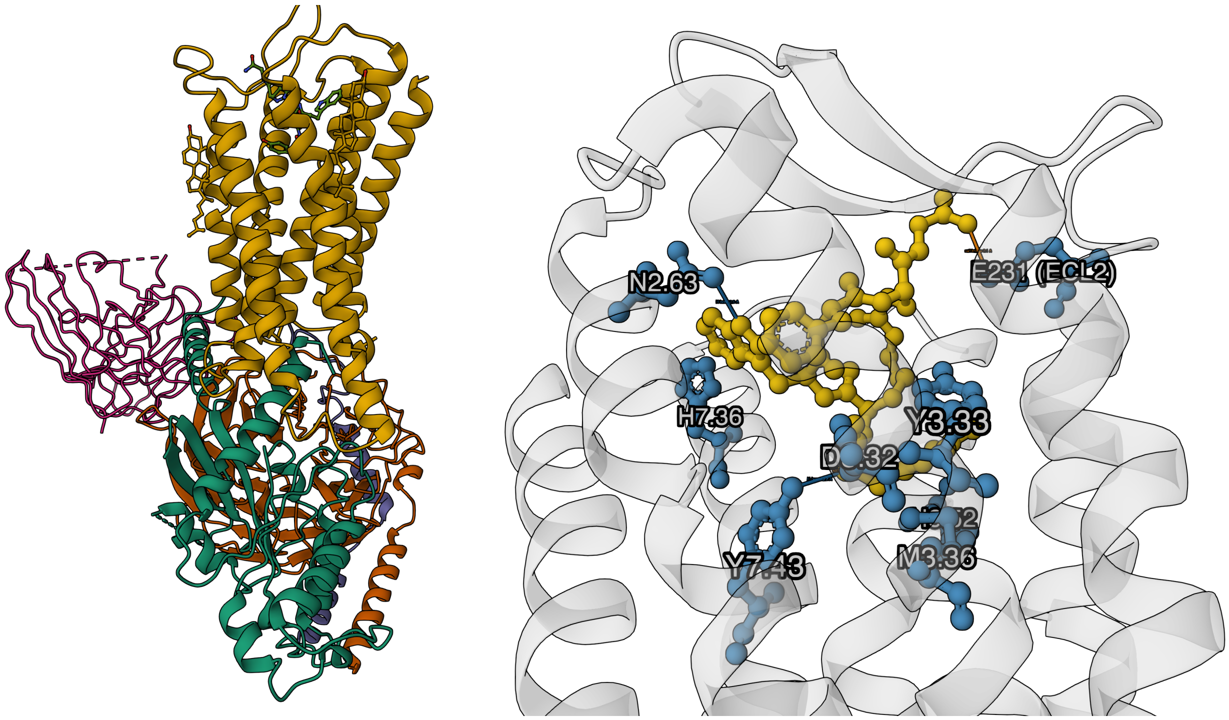
Status: Results for Objectives 1–3 are complete; the molecular-dynamics and free-energy validation (Methods §2.5) is prepared but not yet run — claims dependent on it are flagged [**prospective**].

Abstract

The cyclic endomorphin-1 analog ZH853 is a potent µ-opioid receptor (MOR) agonist with an attractive preclinical profile, but the structural basis of its binding — and how it differs from other opioid agonists — has not been characterized. Using a 3.5 Å cryo-EM structure of the MOR–ZH853–Gi–scFv16 complex, we computed heavy-atom interaction fingerprints for ZH853 and benchmarked them against thirteen deposited MOR complexes spanning peptide agonists (DAMGO, endomorphin-1, β-endorphin), morphinan and fentanyl-class small molecules, biased agonists, and an antagonist. ZH853 engages the canonical opioid anchor (a salt bridge to Asp3.32 satisfied by its Tyr1 α-amine, plus the Tyr7.43 "3–7 lock") shared by all agonists, but additionally forms a **salt bridge to Glu231 in extracellular loop 2 (ECL2) that no other agonist in the set makes**, together with an aromatic/cation-π contact to His7.36 shared only with its endomorphin-1 parent. This extended ECL2/ECL3 engagement is the structural signature of the cyclic scaffold. We nominate **E231Q/E231A** as mutations predicted to disrupt ZH853 selectively while sparing related full agonists, and identify the molecule's beyond-rule-of-5 liabilities (TPSA 235–280 Å^2, 8–10 H-bond donors), proposing two orthogonal optimization series — backbone N-methylation for passive permeability and semaglutide-style C-terminal lipidation for half-life — targeted to solvent-exposed positions established from the bound pose. A membrane molecular-dynamics and free-energy protocol to test these hypotheses is established and released. This is, to our knowledge, the first structural analysis of a cyclic-peptide MOR complex.

1 Introduction

Opioid analgesics acting at the µ-opioid receptor (MOR, gene *OPRM1*) remain the mainstay of severe pain management, but their utility is limited by respiratory depression, tolerance, and dependence. A central goal of contemporary opioid pharmacology is to retain analgesic efficacy while separating it from these liabilities — through signaling bias, partial agonism, or novel chemotypes. Endomorphins, the endogenous MOR-selective tetrapeptides (Tyr-Pro-Trp-Phe-NH_2), are attractive scaffolds because of their high selectivity, but their linear form is metabolically labile and poorly bioavailable. The Zadina laboratory addressed this by engineering cyclic, D-amino-acid endomorphin analogs; **ZH853** (Tyr-cyclo[D-Lys-Trp-Phe-Glu]-Gly-NH_2) is a lead from this series with potent antinociception and a reduced side-effect profile in rodents [1].



(A) **(B)**

Figure 1: The experimentally characterized MOR–ZH853–Gi–scFv16 cryo-EM complex (Mol* renders). **(A)** The whole complex in the canonical GPCR orientation — extracellular face up, intracellular G-protein interface down: μ-opioid receptor receptor 7TM bundle (gold, top) with ZH853 (green sticks) in the extracellular orthosteric pocket, coupling to Gαi (orange), Gβ (teal) and Gγ (purple), with the scFv16 fragment (magenta). **(B)** The orthosteric pocket: ZH853 (yellow ball-and-stick) with key receptor residues (blue) as sticks, labeled by Ballesteros-Weinstein number; salt bridges (orange) to Asp3.32 and ECL2 Glu231 and H-bonds (blue) to Tyr7.43 and Asn2.63 are drawn as distance lines, the receptor cartoon faded for context.

Despite this promise, no structure of ZH853 — or of any cyclic-peptide MOR complex — has been reported, leaving open three questions this work addresses computationally from a 3.5 Å cryo-EM structure of the MOR–ZH853–Gi–scFv16 complex (Figure 1): (1) *what, if anything, is structurally distinctive about how ZH853 engages MOR* relative to other agonists; (2) *which receptor mutations would abrogate ZH853 while sparing related full agonists*, providing a pharmacological fingerprint and a route to mechanistic tests; and (3) *how ZH853's drug-like properties could be improved*, drawing on modern peptide-drug strategies. We further establish and release a membrane molecular-dynamics (MD) and free-energy protocol to validate these structure-derived hypotheses.

2 Related work

MOR structural pharmacology. The inactive MOR structure (β-FNA, 4DKL [2]) and the active BU72 complex (5C1M [3]) established the orthosteric pocket and the activation-associated rearrangements of the W6.48 toggle and TM6. Cryo-EM of MOR–Gi complexes then resolved a series of agonists — the peptide DAMGO (6DDE [4]; human 8EFQ) and endomorphin-1/β-endorphin (8F7R/8F7Q [7]), and the small molecules fentanyl, morphine, oliceridine, SR-17018, PZM21 (8EF5/8EF6/8EFB/8EFL/8EFO [5]) and mitragynine pseudoindoxyl/lofentanil (7T2G/7T2H [6]). All share the D3.32 anchor and a conserved hydrophobic subpocket; biased agonists differ mainly in

TM6/ECL2 engagement. Notably, *every* deposited peptide-agonist MOR structure carries a *linear* peptide — ZH853 would be the first cyclic-peptide complex.

Cyclic peptides as drugs. Macrocyclic peptides occupy "beyond-rule-of-5" (bRo5) chemical space [9]; their oral/passive permeability is governed less by static polarity than by conformational shielding of backbone donors ("molecular chameleons" [10]), which N-methylation and cyclization promote (as in cyclosporine). Half-life, by contrast, is engineered through albumin-binding **lipidation** — the strategy that gives semaglutide a ∼1-week half-life via a γGlu-2×AEEA-C18-diacid conjugate [11]. These two axes — permeability and duration — are distinct and are treated separately here.

3 Methods

3.1 Structure and numbering. We used the deposited real-space-refined MOR–Gi–scFv16–ZH853 model (3.5 Å; $\mathrm{CC_{mask}}$ 0.80; 0.0% Ramachandran outliers). Chains: Gαi (A), Gβ1 (B), Gγ2 (C), scFv16 (D), MOR/*OPRM1* residues 69–349 with 3 modeled cholesterols (R), and ZH853 as HETATM ligand L01 (E). The construct uses **human OPRM1 (UniProt P35372) numbering**, verified against all canonical orthosteric positions; mouse-numbered comparators were offset by +2 to this frame. The ligand's 59 heavy atoms and formula ($C_{42}H_{51}N_9O_8$, 810 Da) match the reference SMILES, confirming the monomeric macrocycle. The overall complex and the orthosteric pocket are shown in Figure 1A,B.

3.2 Comparator set. Thirteen structures were retrieved from the RCSB PDB: peptide agonists 6DDE/8EFQ (DAMGO), 8F7R (endomorphin-1), 8F7Q (β-endorphin), 9WST/9WSV (DAMGO with Gz/β-arrestin); small molecules 5C1M (BU72), 8EF5 (fentanyl), 8EFB (oliceridine), 8EFL (SR-17018), 8EFO (PZM21), 7T2G (mitragynine pseudoindoxyl); and antagonist 4DKL (β-FNA). For each, the receptor chain and bound agonist were isolated programmatically and mapped to human OPRM1 numbering.

3.3 Interaction fingerprints. Because the 3.5 Å model lacks hydrogens, we used a resolution-matched **heavy-atom geometric fingerprint** (rather than angle-based H-bond criteria), classifying each receptor–ligand residue contact as ionic (opposite-charge groups ≤4.0 Å), H-bond (polar N/O ≤3.5 Å), hydrophobic (C–C ≤4.0 Å), π-stacking (aromatic centroids ≤5.5 Å), or cation-π (≤6.0 Å). Aromatic rings of non-standard ligands were detected by planar-ring geometry, a template-free method that generalizes across chemotypes. The same method was applied uniformly to all fourteen complexes.

3.4 Mutation panel and cheminformatics. Candidate selectivity mutations were scored by ZH853 interaction strength × rarity across the twelve agonists × a bonus for sparing the two closest full agonists (endomorphin-1, DAMGO). Physicochemical descriptors (2D, plus 3D radius of gyration, polar SASA fraction, and intramolecular H-bond count from macrocycle-aware conformers) were computed with RDKit; modification variants (N-methylation, lipid conjugation, halogenation) were enumerated programmatically and their properties predicted. Buried vs solvent-exposed ligand atoms were determined from the bound pose (≤4.5 Å to receptor).

The molecular-dynamics protocol below is prepared and released but not yet run [**prospective**]; the design choices and their justifications are given in full so the simulations are reproducible and the modeling assumptions explicit.

3.5 Choice of simulated systems. We define two systems. *System A* is the complete MOR + ZH853 + Gi($\alpha\beta\gamma$) assembly in an explicit bilayer; *System B* is MOR + ZH853 alone with the intracellular half of the receptor Cα-restrained (or the Gα C-terminal α5 helix retained). The rationale is that the active receptor conformation ZH853 stabilizes is thermodynamically coupled to transducer engagement: removing the G protein entirely lets the cytoplasmic end of the 7TM bundle (notably TM6) relax toward inactive rotamers on accessible MD timescales, corrupting the very orthosteric geometry we set out to characterize [12]. Retaining the heterotrimer (System A) holds the active state *without* artificial restraints and exposes allosteric coupling, at the cost of a large (∼200,000-atom) system; System B trades that fidelity for the throughput the free-energy calculations require, using minimal restraints to preserve activation. The scFv16 fragment is a cryo-EM chaperone of the Gαi–Gβ interface with no physiological role and is removed so it cannot bias the dynamics. Because the D2.50 sodium-pocket aspartate lies at the activation-linked allosteric site [13] and its protonation is ambiguous at physiological pH (§3.6), every system is built in both D2.50 protonation states.

3.6 Receptor and ligand preparation. At 3.5 Å many surface sidechains are under-determined; we rebuild the 45 truncated sidechains with PDBFixer [14] rather than leaving partial residues that would open spurious cavities or drop formal charges. Protonation is assigned by PROPKA at pH 7.4 [15]; the sole non-standard prediction is **D2.50 (Asp116), pKa 7.61** — essentially 1:1 protonated at physiological pH — motivating the parallel-system design and constant-pH MD as the rigorous fallback. All four chain-R histidines are neutral; their HID/HIE tautomers are set from the local hydrogen-bond network because H6.52 and H7.36 line the pocket and a mis-assigned tautomer would perturb the interactions under study. The conserved ECL2–TM3 disulfide (Cys142–Cys219) is enforced explicitly, and the truncated receptor termini are capped with neutral acetyl/*N*-methylamide groups rather than charged termini, which would be artifacts on an internal transmembrane fragment. ZH853 is a macrocyclic peptide with D-amino acids and a side-chain-to-side-chain lactam — outside the validated coverage of both general small-molecule force fields (not optimized for peptides) and protein force fields (which lack the non-canonical linkage). We therefore parameterize it by an Amber residue-library route: capped-dipeptide fragments per residue, multi-conformer RESP charges at HF/6-31G(d) [16] to match the derivation of the ff19SB protein charges it must interface with, GAFF2 [17] terms for novel side-chain chemistry, explicit D-chirality, and head-to-tail macrocycle closure; a whole-molecule GAFF2/AM1-BCC build and an independent OpenFF build serve as cross-checks. The physiologically dominant +1 state is used (the Tyr1 α-amine that forms the D3.32 salt bridge; there is no free carboxylate). Because fixed-charge force fields reproduce macrocyclic *cis*/*trans* amide equilibria and ring-pucker populations poorly [18], production is augmented with enhanced sampling and, where data exist, validated against NMR.

3.7 Membrane placement and force field. The receptor is placed in the bilayer by the standard **OPM/PPM transfer-energy method** [19, 29]. Because our cryo-EM model is not itself in OPM, we transfer the placement structurally: our receptor is superposed (Cα Kabsch, RMSD 1.1 Å) onto the OPM-oriented MOR–Gi–DAMGO reference (6DDF), whose membrane boundaries (DUM pseudo-atoms) give a **hydrophobic thickness of 31.4 Å** — consistent with OPM across active MOR structures (31–35 Å; 4DKL 32.0±1.0 Å) and adopted as the build target. We do *not* derive the membrane from the resolved lipids: the 3 cholesterols bind at site-specific motifs, each spans about one leaflet, and 3 molecules fix the midplane only to ∼2 Å. Two features of our own map corroborate the OPM placement (Figure 2): the **Trp/Tyr aromatic girdle** — interfacial residues that snorkel to the headgroup regions [30] — forms two bands ∼30 Å apart (interfaces at $z \approx +13$ and -17 Å), and the cholesterols span ∼33 Å about the OPM midplane; both bracket the 31.4 Å slab. This thickness matches the experimental POPC hydrocarbon core ($2D_c$=28.8 Å[31]) plus the cholesterol condensing effect (+2–4 Å for a POPC-like bilayer), within a phosphate-to-phosphate span of ∼37–40 Å. For a definitive per-structure value the model can be submitted to the PPM 3.0 server. The receptor is embedded in a POPC:cholesterol (9:1) membrane using CHARMM-GUI Membrane Builder [20] — a widely used minimal GPCR bilayer that captures cholesterol's documented modulation of class-A receptors [21] while remaining tractable; a higher-cholesterol asymmetric plasma-membrane mimic is run as a sensitivity check since MOR is cholesterol-sensitive. The Gi heterotrimer projects into the cytoplasmic water slab, so extra intracellular padding is added to prevent periodic-image self-interaction of the large soluble domain. We use **ff19SB + Lipid21 + OPC** [22, 23, 24] with 0.15 M NaCl: ff19SB's residue-specific CMAP corrections improve backbone conformational balance and are calibrated with OPC water, so we deliberately avoid mismatching the water model to the protein force field — a common, consequential error (CHARMM36m/CHARMM-GUI [25] is the validated alternative). Equilibration follows a six-stage schedule with positional restraints on protein and lipid heavy atoms released gradually (backbone 4000→10, sidechain 2000→0, lipid 1000→0 $\mathrm{kJ\,mol^{-1}\,nm^{-2}}$) across NVT then NPT stages, letting lipids pack around the receptor and water reorganize before the solute is freed; abrupt restraint release is a known source of membrane artifacts.

3.8 Production and quality control. Production (OpenMM [26]) uses a LangevinMiddle integrator at 310 K with a MonteCarloMembraneBarostat (semi-isotropic, zero surface tension) so the bilayer area relaxes naturally, and hydrogen-mass repartitioning to enable a **4 fs** timestep [27], roughly doubling throughput with no measurable cost to the observables of interest. We run **≥3 independent replicas** of several hundred nanoseconds each, because convergence is judged from the agreement of independent trajectories and block-averaged errors rather than a single RMSD plateau, which can flatten while the ensemble is still unconverged [28]. Reported metrics are backbone Cα RMSD/RMSF (aligned before RMSF), *receptor-aligned* ligand RMSD (so it reports pocket escape rather than global drift), key-contact occupancy (D3.32, ECL2 Glu231, H6.52, H7.36, Y7.43), and membrane observables (area-per-lipid ≈0.65–0.68 nm^2 for POPC, bilayer thickness, deuterium order parameters) benchmarked against experiment to validate the bilayer. Crucially, the contact occupancies convert the static census of Figure 5 into equilibrium probabilities, directly testing whether the ECL2 Glu231 salt bridge that distinguishes ZH853 is persistent or transient.

3.9 Free-energy calculations. Relative FEP across the ZH850/831/809 analogs (with ABFE and funnel metadynamics as exploratory) is released as a SLURM bundle. Given documented convergence difficulties for flexible macrocycles, these are framed as rank-ordering rather than quantitative affinities.

All analyses are scripted and version-controlled (Data & code availability).

4 Results

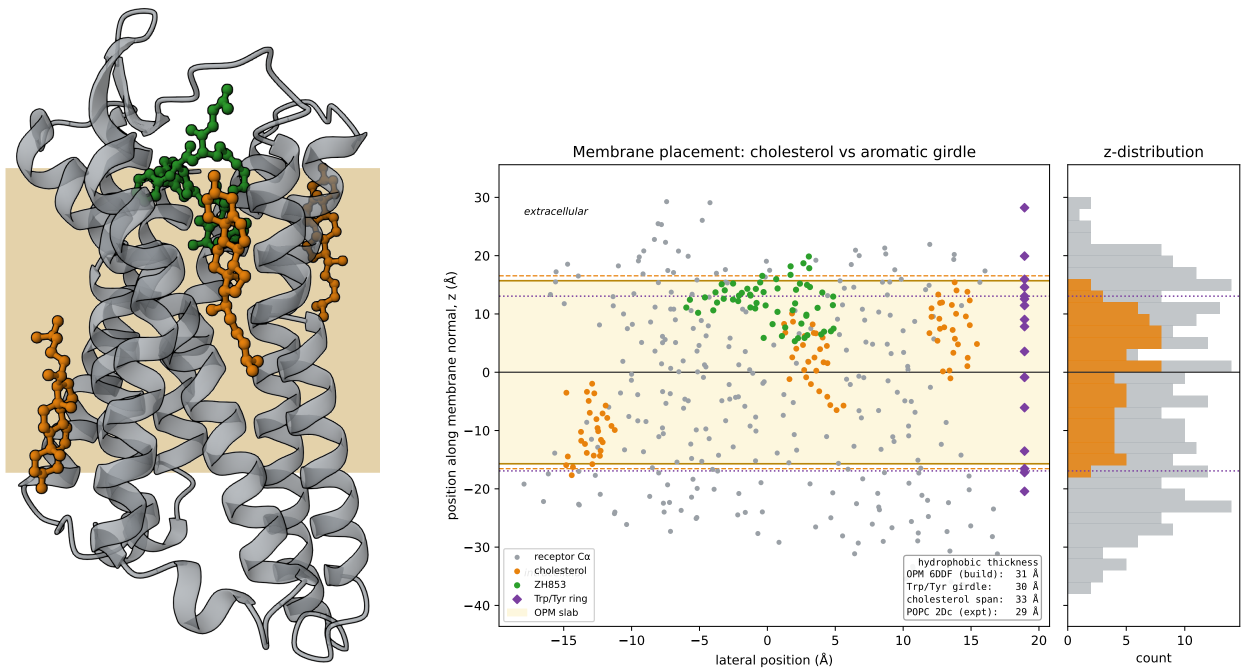
4.1 ZH853 satisfies the conserved opioid anchor

ZH853 contacts 25 receptor residues. Its Tyr1 α-amine forms the **canonical Asp3.32 (Asp149) salt bridge** (3.16 Å) and hydrogen-bonds Tyr7.43 (Tyr328, 3.17 Å) — the "3–7 lock" — while the Tyr1 ring and the macrocyclic Trp/Phe pack into the conserved hydrophobic subpocket (Met3.36, Val3.28/Ile3.29, Ile6.51, Val6.55, Val5.42; Figure 4). Every one of these contacts is shared by all twelve comparator agonists (Figure 5), confirming that ZH853 presents the same "message" pharmacophore as morphine, fentanyl, DAMGO, and endomorphin-1. Consistent with this, PROPKA assigns Asp149 a depressed pKa (6.0), i.e. charged and salt-bridge-competent.

4.2 A distinctive ECL2 salt bridge distinguishes ZH853 (Objective 1)

Against this conserved background, three contacts set ZH853 apart (Figure 5; Table 1):

- **Glu231 (ECL2): an ionic + H-bond contact made by ZH853 alone (0/12 agonists).** A ZH853 basic group reaches ECL2 to form a salt bridge (3.12 Å) that no linear peptide or



(A) **(B)**

Figure 2: Membrane placement and thickness validation. **(A)** Side view (Mol* render) of the oriented µ-opioid receptor (grey cartoon) with the 3 modeled cholesterols (84 atoms; orange, ball-and-stick) and ZH853 (green). The **tan band is the inferred hydrophobic core** (§3.7), drawn behind the receptor: the transmembrane helices span it and the cholesterols sit within it, while the intracellular loops fall below and ZH853 sits at the extracellular boundary. **(B)** Placement cross-check along the membrane normal z: receptor Cα (grey), cholesterols (orange), ZH853 (green), and the Trp/Tyr aromatic girdle (purple diamonds; interfaces dotted). The **OPM slab** (yellow band, 31.4 Å, from the 6DDF superposition) is bracketed by the aromatic girdle ($\sim$30 Å) and the cholesterol span ($\sim$33 Å, dashed), and matches the experimental POPC hydrocarbon core ($2D_c$=28.8 Å); ZH853 localizes to the extracellular leaflet, confirming the orientation.

small molecule in the set reproduces. PROPKA confirms Glu231 is charged (pKa 4.9). This is the single strongest structural discriminator.

- **His7.36 (His321): aromatic/cation-π, shared only with endomorphin-1 and mitragynine (2/12).** An ECL3-proximal contact retained from the endomorphin-1 lineage.
- Weaker ZH853-unique contacts at Leu221/Phe223 (ECL2) and Leu234 (TM5), gained by the extended/cyclic scaffold.

The picture is coherent: **ZH853 keeps the deep conserved anchor but reaches "up and out" toward ECL2/ECL3**, engaging a rim region that the compact linear agonists do not. This extended loop engagement is the structural hallmark of the macrocycle.

4.3 A selectivity handle: E231 mutations (Objective 2)

Ranking pocket residues by ZH853 interaction strength, rarity across agonists, and sparing of the endomorphin-1/DAMGO parents nominates **Glu231 as the top selective target** (Table 1). We predict **E231Q** (charge-neutral isostere) and **E231A** (removal) will attenuate ZH853 binding while leaving agonists that do not contact ECL2 Glu231 largely unaffected — a testable pharmacological signature. His7.36 substitutions (H321A/F) are a secondary, less clean handle (they also perturb endomorphin-1). By contrast, the universal anchors (Asp3.32, Tyr7.43, Tyr3.33, Met3.36, and the hydrophobic cage) are **do-not-mutate controls**: disrupting them abrogates all agonists and cannot confer selectivity. [**prospective**] magnitudes await MD occupancy and relative-FEP/in-silico-mutagenesis validation.

4.4 Drug-likeness liabilities and a two-axis design strategy (Objective 3)

All four analogs (Figure 3) sit firmly beyond rule-of-5 (MW 714–810, TPSA 235–280 $\mathrm{\AA}^2$, 8–10 H-bond donors; Figure 6); the dominant liabilities are high polar surface area and donor count. Their 3D conformers show 4–6 intramolecular H-bonds — partial donor self-shielding consistent with chameleonic behavior. From the bound pose, **46 of 59 ligand heavy atoms are buried** (the Tyr1/aromatic pharmacophore) and **13 are solvent-exposed**, defining where derivatization is structurally safe. We therefore propose two orthogonal series (Figure 7):

1. **Permeability** — backbone **N-methylation** of solvent-facing amides (each removes one donor and $\sim$9 $\mathrm{\AA}^2$ TPSA; a 2–3-site subset reaches TPSA $\approx$253 $\mathrm{\AA}^2$, crossing the oral-bRo5 threshold), optionally with 4-F-Phe for metabolic stability. Best pursued from **ZH831**, the least liable analog.
2. **Half-life** — **C-terminal semaglutide-style lipidation** on the exposed cap (γGlu-2$\times$AEEA-C18-diacid), trading polarity for albumin-mediated duration; pursued from the most potent analog, **ZH853**.

Both series must be potency-checked against MOR by relative FEP, and N-methyl sites chosen to avoid the receptor-facing backbone amides identified in §4.1–4.2.

5 Discussion

The central finding is that ZH853's distinctiveness lies not in the conserved orthosteric anchor — which it shares with all opioid agonists — but in a **cyclic-scaffold-enabled ECL2 salt bridge**

(Glu231) and adjacent ECL3 aromatic contact (His7.36). ECL2 is a recognized determinant of ligand selectivity and kinetics across GPCRs, and its engagement here provides a plausible structural correlate for ZH853's distinct pharmacology and a concrete, low-ambiguity selectivity handle (E231Q). Because Glu231 is peripheral to the conserved message, its mutation is well suited to a clean pharmacological separation of ZH853 from morphinan/fentanyl chemotypes and even from its linear endomorphin-1 parent.

The design analysis reframes ZH853 optimization as two independent problems. The molecule already banks the hard-won macrocyclic and D-amino-acid protease resistance; what remains is permeability (a donor/PSA problem, addressable by N-methylation guided by which amides face solvent vs receptor) and duration (a half-life problem, addressable by lipidation on the exposed cap). Coupling the design to the bound pose avoids the common error of derivatizing the pharmacophore.

Limitations. All results derive from a single 3.5 Å static model; sidechain rotamers and the exact ligand pose carry real uncertainty, and interaction assignments use heavy-atom geometry rather than explicit hydrogens. The comparative analysis is a contact census, not an energetic ranking. The mutation and design predictions are hypotheses; their magnitudes require the MD/FEP validation whose protocol we establish here, and macrocycle free-energy convergence is itself a known challenge.

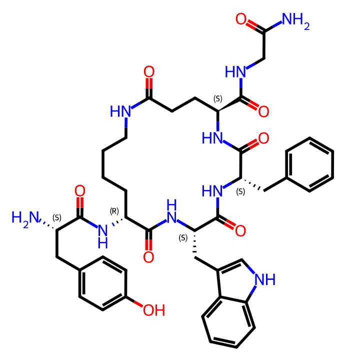
6 Conclusions and future work

From the first structural analysis of a cyclic-peptide MOR complex, we identify an ECL2 Glu231 salt bridge as the defining, ZH853-specific interaction, nominate E231Q/E231A as selectivity-conferring mutations, and propose bound-pose-guided N-methylation and lipidation series to address the molecule's permeability and half-life liabilities. The immediate next step is to run the released membrane-MD and free-energy protocol to (i) confirm contact occupancy and pose stability, (ii) quantify the E231 selectivity, and (iii) rank the analog and designed-variant affinities. Longer term, the Gz and β-arrestin comparators included here support a follow-on analysis of whether ECL2/ECL3 engagement correlates with ZH853's reported signaling bias.

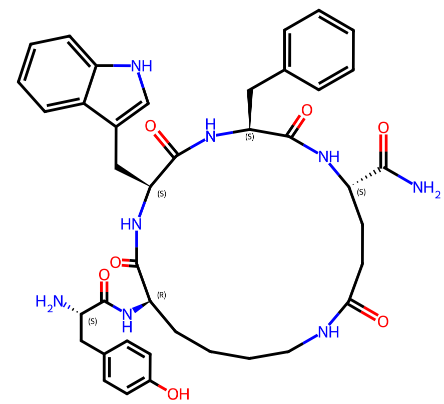
Figures

Table 1: ZH853 contacts ranked by distinctiveness (excerpt). Full interaction matrix and mutation panel are provided as dated `product/` outputs (Data and code availability).

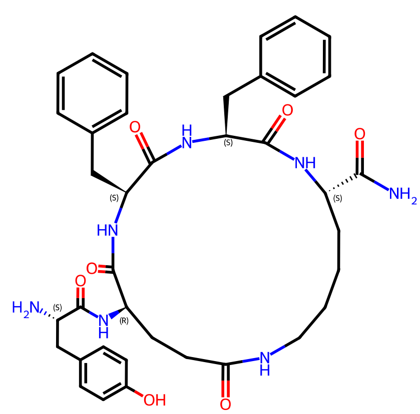
Residue	BW	ZH853 interaction	#/12	Note
Glu231	ECL2	ionic + H-bond	**0**	ZH853-unique; primary selectivity handle (E231Q/E231A)
His321	7.36	π-stack + cation-π	2	shared only with endomorphin-1, mitragynine
Tyr130	2.64	hydrophobic	1	endomorphin-1 only
Leu221/Phe223	45.52/45.54	contact	0	weak ECL2 contacts (cyclic scaffold)
Asp149	3.32	ionic + H-bond	12	universal anchor — do-not-mutate control
Tyr328	7.43	H-bond	12	universal (3–7 lock) — control



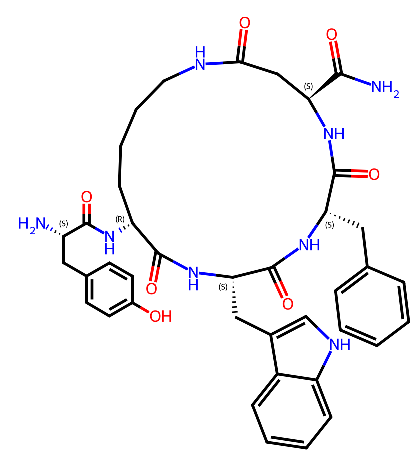
ZH853 (810 Da)



ZH850 (753 Da)



ZH831 (714 Da)



ZH809 (739 Da)

Figure 3: ZH853 and the three FEP analogs (ZH850/831/809), drawn as scaffold-aligned 2D structures with standard atom coloring and CIP stereo descriptors. All are Tyr-cyclo[...]-NH_2 endomorphin-1 macrocycles differing by single-residue substitutions (Trp↔Phe, Glu↔Asp ring size, ±Gly cap).

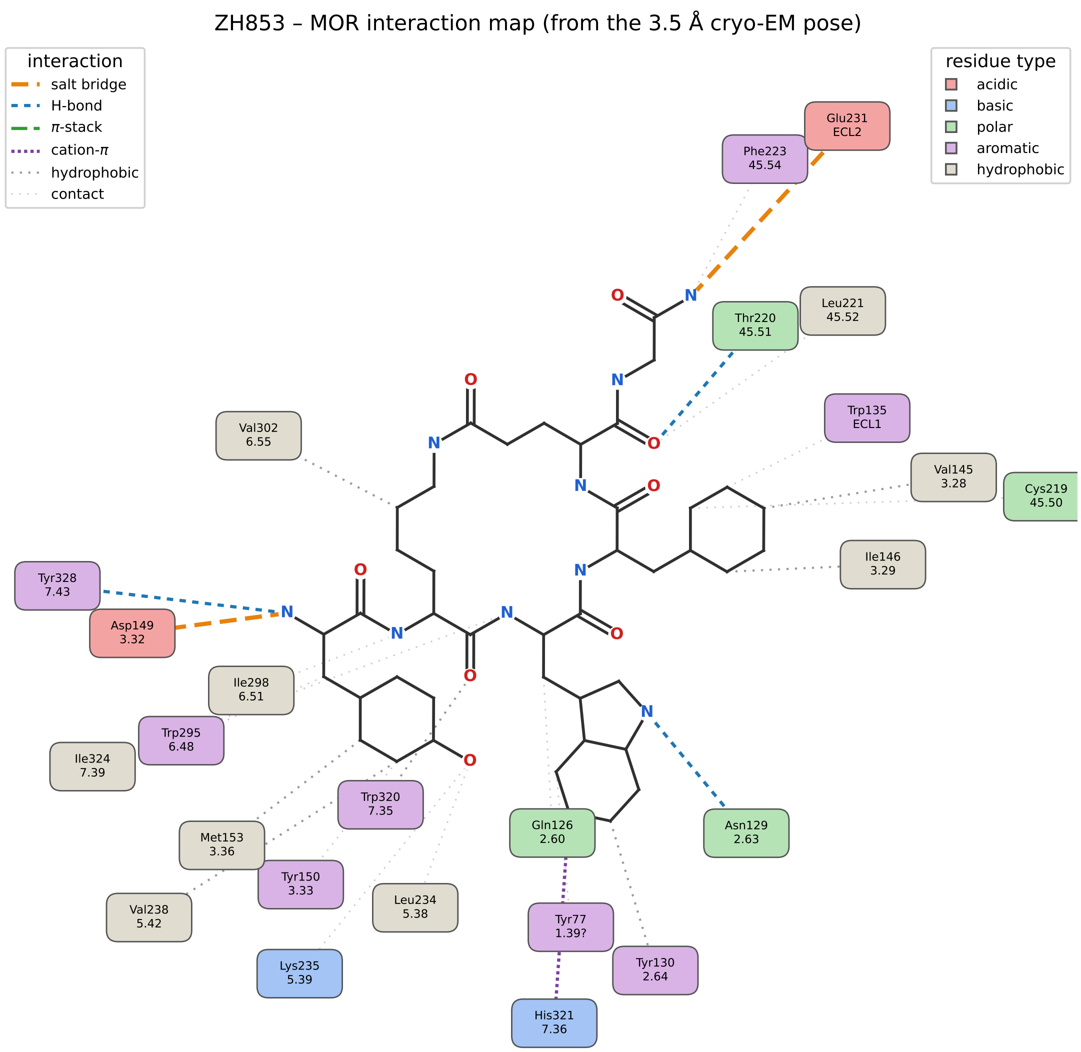


Figure 4: PoseView-style 2D interaction map of ZH853 in the MOR pocket, generated from the 3.5 Å cryo-EM coordinates. The ligand skeleton (atom-colored) is surrounded by contacting residues, colored by chemical class and placed in the direction of their contact atom; connectors are styled by interaction type. The Asp3.32 (Asp149) and ECL2 Glu231 salt bridges (orange) and the Tyr7.43/Asn2.63 H-bonds (blue) are highlighted.

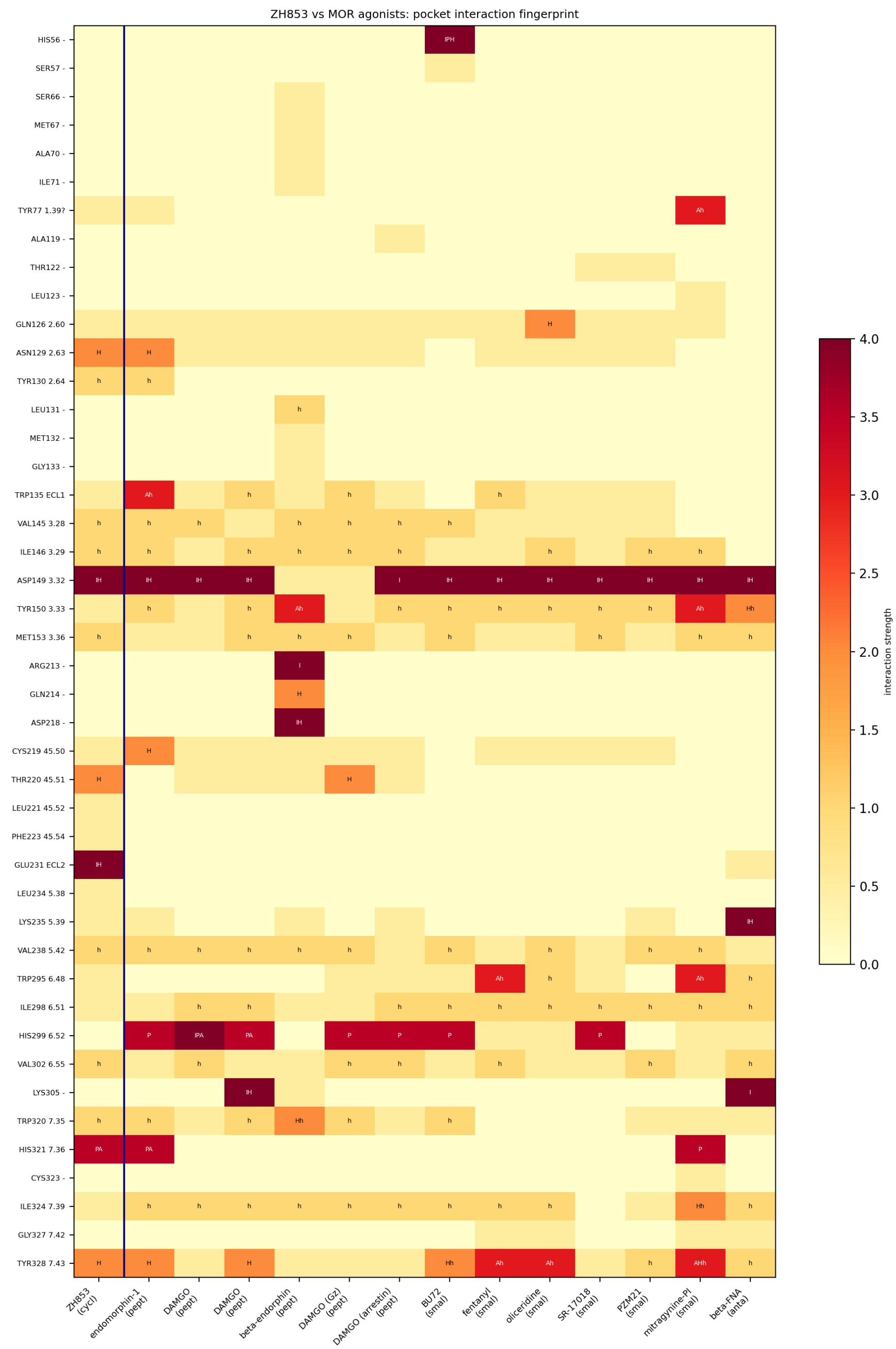


Figure 5: Pocket interaction fingerprint of ZH853 vs 13 MOR complexes. Rows = receptor residues (human OPRM1 / Ballesteros-Weinstein); columns = complexes; cells colored by interaction strength and labeled by type (I ionic, P cation-π, A π-stack, H H-bond, h hydrophobic). Asp3.32 is engaged across all columns; Glu231 (ECL2) is engaged in the ZH853 column alone.

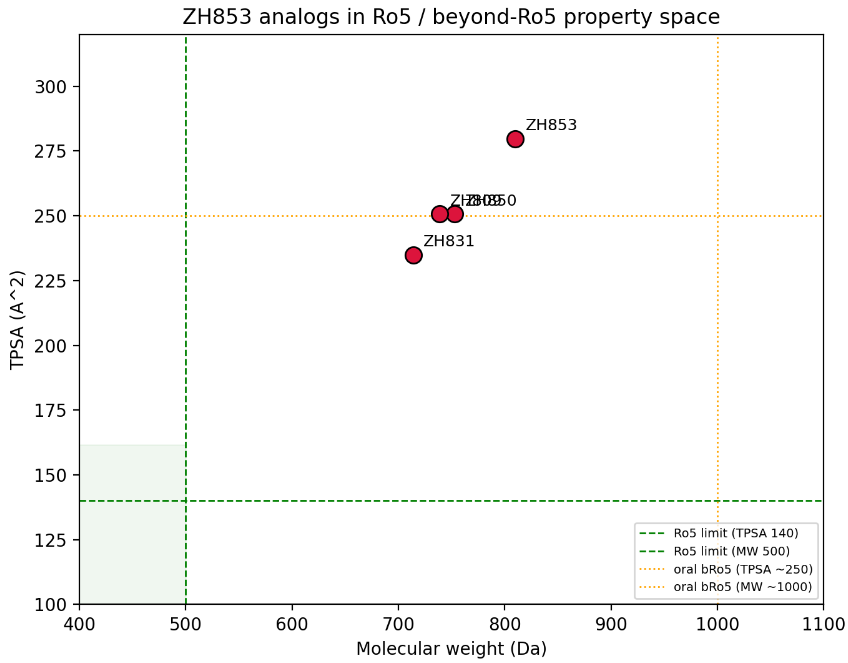


Figure 6: ZH853 analogs in Ro5 / beyond-Ro5 property space (MW vs TPSA), with Lipinski and oral-bRo5 reference limits.

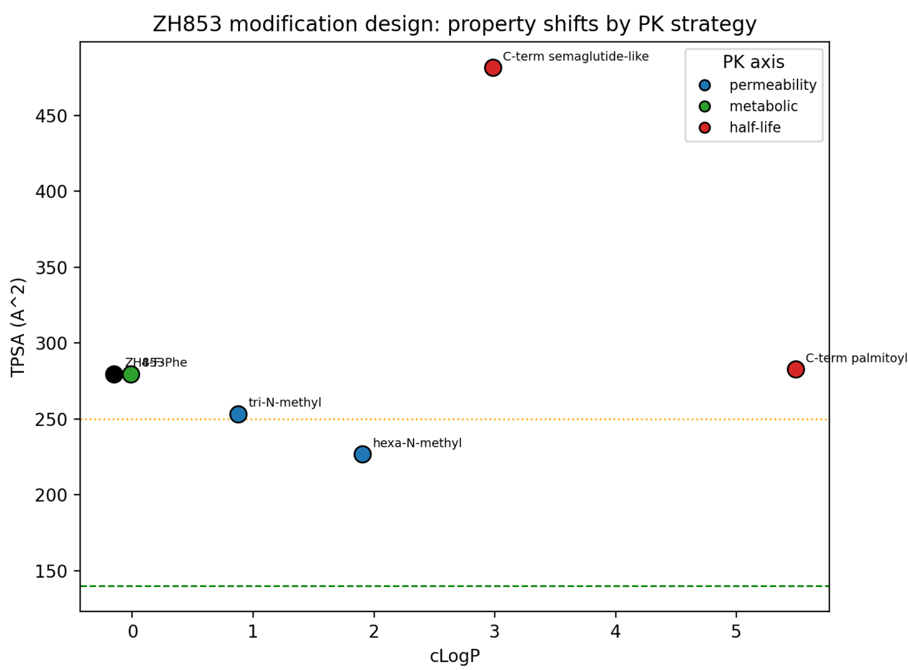


Figure 7: Predicted property shifts of the modification strategies by PK axis (TPSA vs cLogP): N-methylation (permeability) lowers TPSA across the bRo5 line; lipidation (half-life) raises lipophilicity.

Data and code availability

All analyses are scripted and version-controlled in the project repository (`src/`, numbered workflows; `zh853mor` package; `make analysis` reproduces every figure and table). Comparator structures are re-fetchable via `make fetch`. Decisions and clarifications are logged in `SPECIFICATION.md`; detailed methods and results are in the `docs/` directory (`METHODS_md_prep`, `RESULTS_interactions`, `RESULTS_analog_design`).

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